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Machine learning-based prediction of friction stir welding parameters for optimized tensile strength in aluminum alloys

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Abstract

Friction stir welding (FSW) is a widely adopted method in industries such as aerospace, automotive, and marine for joining aluminum alloys, offering enhanced weld quality and mechanical properties. However, the optimization of FSW parameters remains a complex

task due to the intricate relationships between process variables and weld characteristics. This study aims to develop a predictive model for estimating key FSW parameters, such as rotation speed, welding speed, and tilt angle, using machine learning techniques. An Auto-Encoder model is proposed and compared against traditional models including Long Short-Term Memory (LSTM), Support Vector Machines (SVM), and Gaussian Process Regression (GPR). The dataset includes FSW experiments on similar (AA7075) and dissimilar (AA5083 and AA6061) aluminum alloys. The Auto-Encoder model consistently outperforms other methods in predicting tensile strength, with a 5.679% average error for AA7075, demonstrating its superior accuracy and potential for industrial applications. The findings provide a robust framework for optimizing FSW processes, significantly reducing the trial-and-error approach traditionally used in parameter estimation.

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Conceptualization, writing—original draft preparation, PL.; review, funding acquisition, SK.; methodology, SG.; validation, writing review and editing, SC.; data curation, SM.; writing review and editing, WS.; supervision, project administration, TS.; visualization GJ. All authors have read and agreed to the published version of the manuscript.

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